

Design and Implementation of a Small Virtual Client-Server Network Using Cisco Packet Tracer

FINAL PROJECT REPORT

Course: CSCE 3530 - Computer Networks

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Platform: Cisco Packet Tracer

Project Scope

A functional small-company virtual network demonstrating DHCP, DNS, ARP, HTTP, HTTPS/SSL-TLS, ICMP, TCP, UDP, NAT/PAT, IP addressing, routing, and packet analysis.

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1. Executive Summary

This project designed, configured, tested, and documented a small virtual client-server network in Cisco Packet Tracer. The network represents a small-company environment with two internal client computers, one internal multi-service server, a Layer 2 switch, a router, and an external server used to simulate internet communication.

The completed environment successfully demonstrated automatic address assignment through DHCP, internal hostname resolution through DNS, web communication through HTTP and HTTPS, connectivity testing through ICMP, transport-layer behavior through TCP and UDP, MAC-address discovery through ARP, routing between separate IPv4 networks, and private-to-public address translation through NAT overload (PAT). Packet Tracer Simulation Mode was used to inspect protocol behavior at multiple OSI layers.

Overall Result

The network was fully functional. Both client PCs received DHCP addresses, resolved `www.company.local`, loaded the internal web page over HTTP and HTTPS, reached the external server, and generated visible DNS, DHCP, ICMP, TCP, HTTP, and NAT evidence.

2. Project Objectives and Scope

- Create a working client-server network using virtual routers, switches, PCs, and servers.
- Configure correct IPv4 addressing for internal and external network segments.
- Use DHCP to assign client addresses automatically.
- Use DNS to resolve `www.company.local` to the internal web server.
- Demonstrate HTTP and HTTPS client-server communication.
- Validate connectivity using ping and traceroute.
- Observe ARP, ICMP, DNS, DHCP, TCP, UDP, and HTTP traffic in Simulation Mode.
- Configure NAT overload so private internal clients can communicate with an external network.
- Document configuration steps, testing results, challenges, and solutions.

3. Tools and Technologies

Tool / Technology	Purpose
Cisco Packet Tracer	Network design, device configuration, simulation, and packet inspection.
Cisco 2911 Router	Inter-network routing and NAT/PAT.
Cisco 2960 Switch	Layer 2 connectivity for the internal LAN.
Server-PT	DHCP, DNS, HTTP, and HTTPS services; external-host simulation.
PC-PT	DHCP clients, web clients, ping, traceroute, ARP, and DNS testing.
Simulation Mode	Protocol filtering and OSI/PDU analysis.
Router CLI	Interface addressing, NAT configuration, and verification commands.

4. Network Design

The logical design contains two IPv4 /24 networks. The internal LAN uses 192.168.1.0/24 and connects the clients and internal server through a Cisco 2960 switch. The router LAN interface is the default gateway. The external segment uses 203.0.113.0/24 and contains the router WAN interface and the external server.

The router separates the two broadcast domains and performs NAT overload for traffic originating from private internal addresses. The internal server combines DHCP, DNS, HTTP, and HTTPS services to represent a small organization with centralized infrastructure.

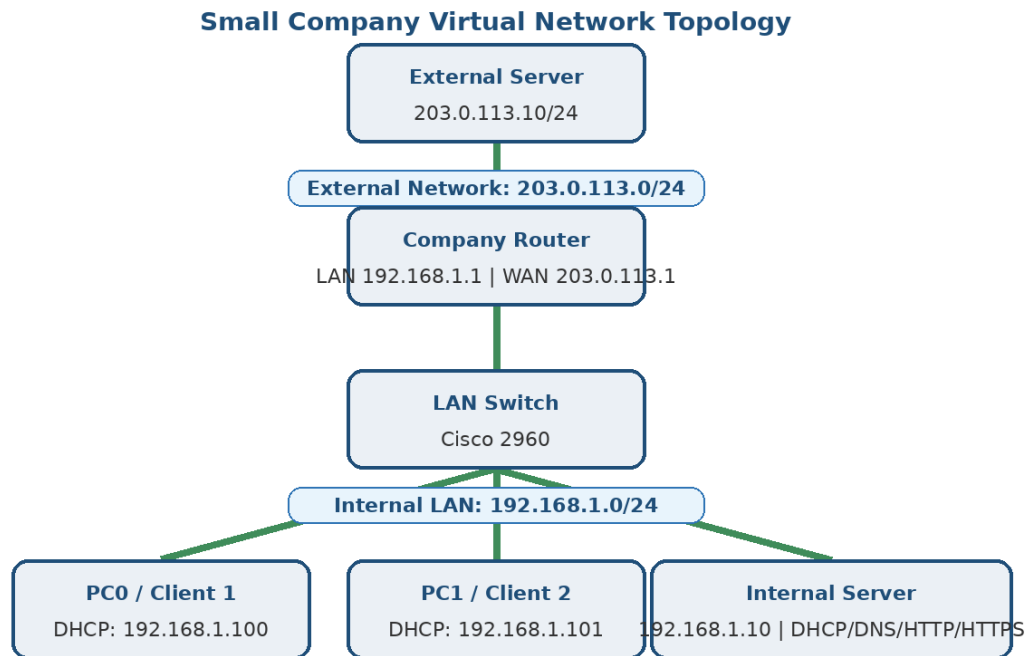
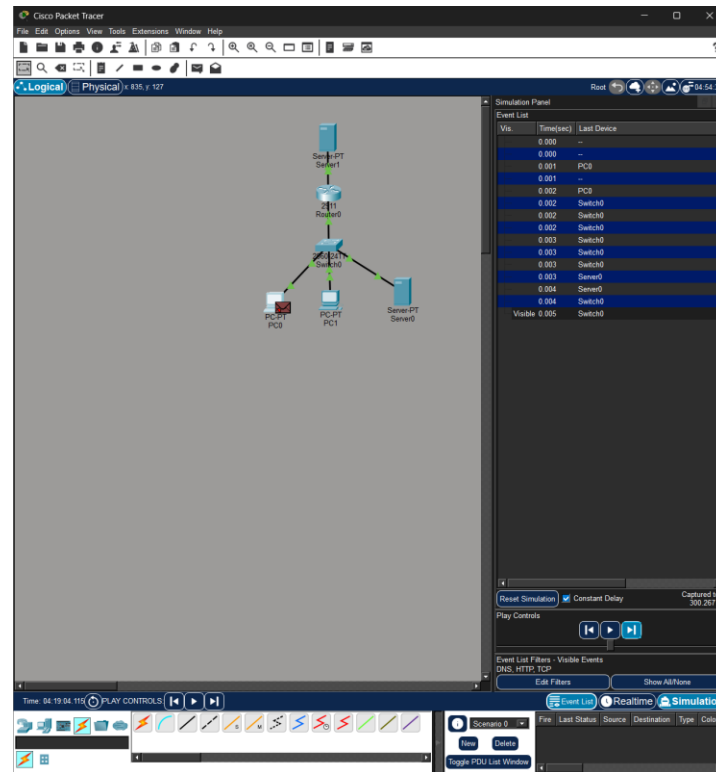


Figure 1. Final Cisco Packet Tracer topology with all physical links operational (green).



5. IP Addressing Plan

Device	Interface / Role	IPv4 Address	Subnet Mask	Default Gateway
Router0	GigabitEthernet0/0 - LAN	192.168.1.1	255.255.255.0	N/A
Router0	GigabitEthernet0/1 - WAN	203.0.113.1	255.255.255.0	N/A
Server0	DHCP / DNS / HTTP / HTTPS	192.168.1.10	255.255.255.0	192.168.1.1
PC0	DHCP client	192.168.1.100	255.255.255.0	192.168.1.1
PC1	DHCP client	192.168.1.101	255.255.255.0	192.168.1.1
Server1	External server	203.0.113.10	255.255.255.0	203.0.113.1

The DHCP pool begins at 192.168.1.100 and supports 51 users, covering 192.168.1.100 through 192.168.1.150. The DNS server address distributed to clients is 192.168.1.10.

6. Device Configuration

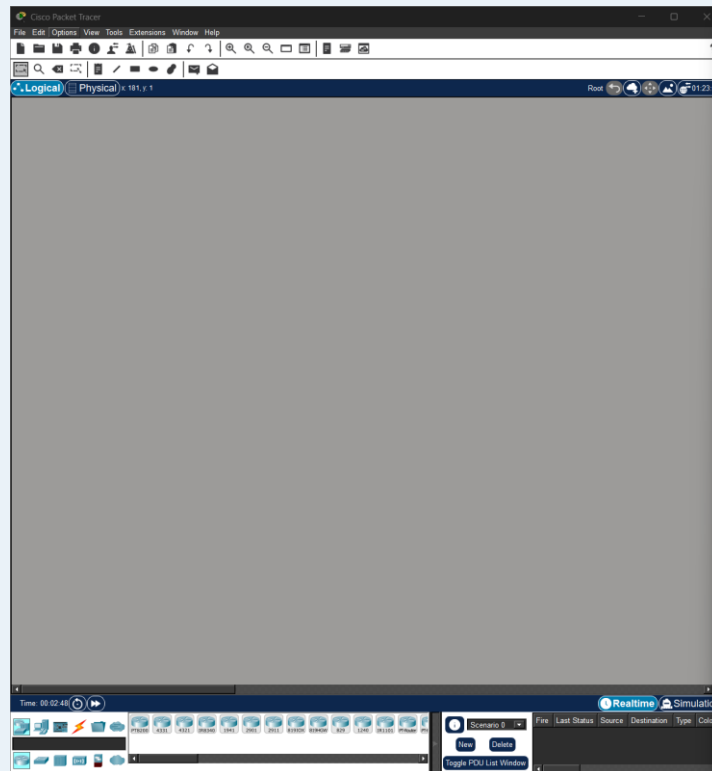
6.1 Router Interfaces

```
enable
configure terminal
interface gigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
 exit
interface gigabitEthernet0/1
 ip address 203.0.113.1 255.255.255.0
 no shutdown
 exit
```

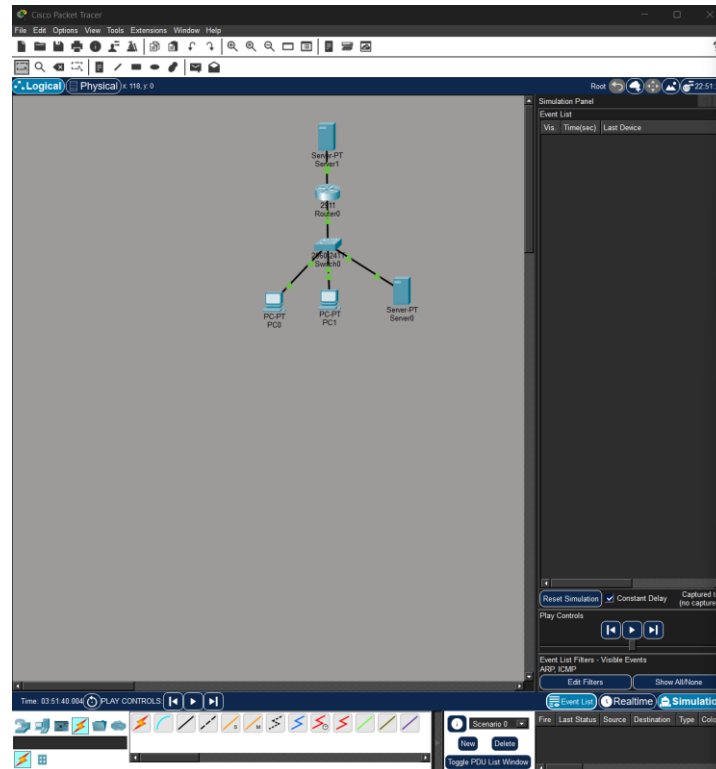
```
end  
write memory
```

Verification

show ip interface brief confirmed GigabitEthernet0/0 = 192.168.1.1 up/up and GigabitEthernet0/1 = 203.0.113.1 up/up.

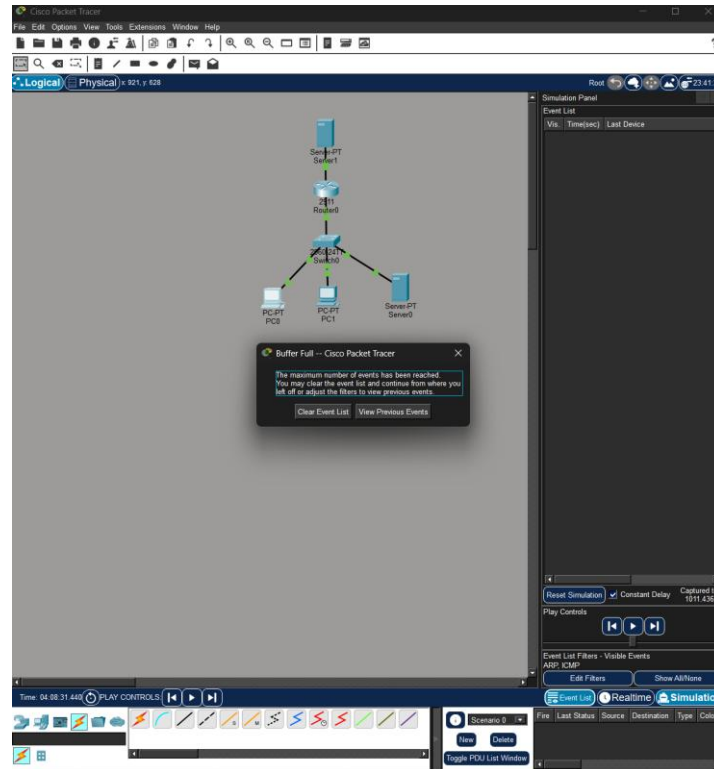


6.2 Internal Server



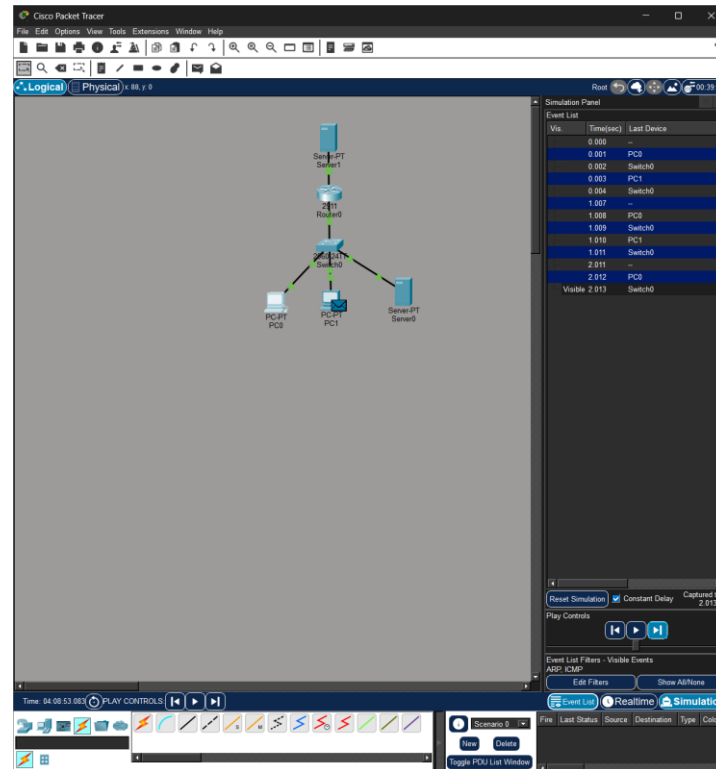
- Static IPv4 address: 192.168.1.10/24
- Default gateway: 192.168.1.1
- DNS server: 192.168.1.10
- DHCP service: enabled
- DNS service: enabled
- HTTP and HTTPS services: enabled

6.3 DHCP Pool



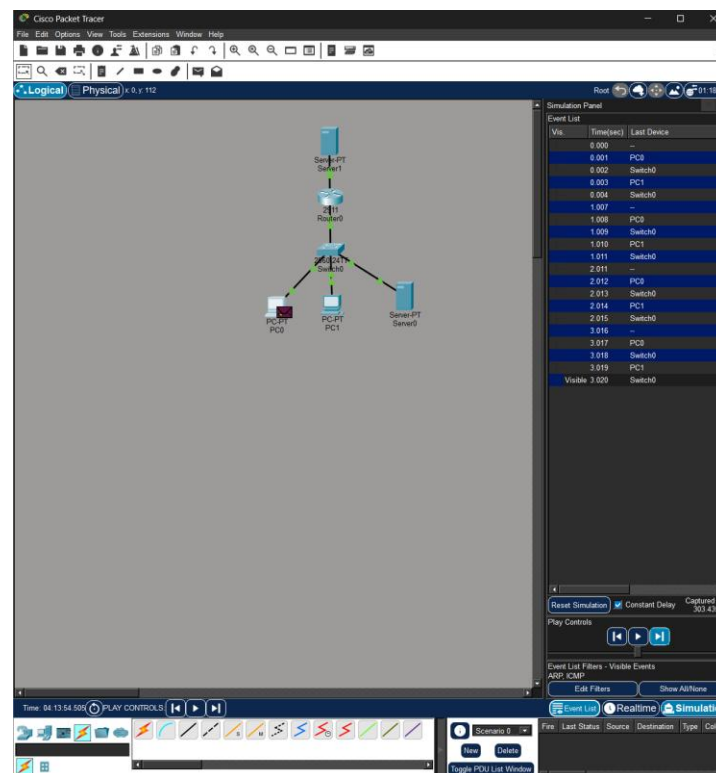
Setting	Value
Pool Name	serverPool / LAN pool
Default Gateway	192.168.1.1
DNS Server	192.168.1.10
Starting Address	192.168.1.100
Subnet Mask	255.255.255.0
Maximum Users	51

6.4 DNS Record



A Record: `www.company.local -> 192.168.1.10`

6.5 Web Server



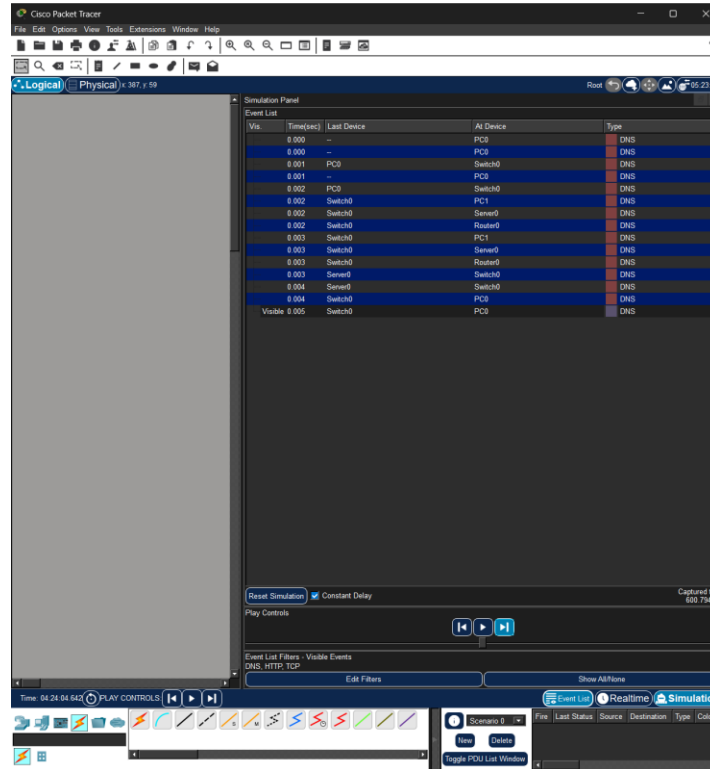
```
<html>
<head><title>CSCE 3530 Network Project</title></head>
```

```

<body>
<h1>CSCE 3530 Company Network</h1>
<p>Welcome to the internal company web server.</p>
<p>DHCP, DNS, HTTP, ICMP, TCP, UDP, and NAT are being demonstrated.</p>
</body>
</html>

```

6.6 NAT/PAT Configuration



```

enable
configure terminal
access-list 1 permit 192.168.1.0 0.0.0.255
interface gigabitEthernet0/0
 ip nat inside
 exit
interface gigabitEthernet0/1
 ip nat outside
 exit
ip nat inside source list 1 interface gigabitEthernet0/1 overload
end
write memory

```

7. Protocol Demonstrations and Analysis

7.1 DHCP and UDP

PC0 and PC1 successfully obtained 192.168.1.100 and 192.168.1.101 from the DHCP server. The DHCP Discover used source IP 0.0.0.0, destination IP 255.255.255.255, UDP source port 68, UDP destination port 67, and destination MAC FFFF.FFFF.FFFF. Server responses used UDP source port 67 and destination port 68. This confirms the DORA process and DHCP broadcast behavior.

PDU Information at Device: Switch0

OSI Model Inbound PDU Details Outbound PDU Details

Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header
00D0.FF33.2BEB >> FFFF.FFFF.FFFF
Layer 1: Port FastEthernet0/2

Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header
00D0.FF33.2BEB >> FFFF.FFFF.FFFF
Layer 1: Port(s): FastEthernet0/1
FastEthernet0/3 GigabitEthernet0/1

1. FastEthernet0/2 receives the frame.

Challenge Me << Previous Layer Next Layer >>

Figure 2. DHCP Discover broadcast observed at Router0: UDP 68 to 67, source 0.0.0.0, destination 255.255.255.255.

```

Router0
Physical Config CLI Attributes

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wvl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400K5
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit

```

7.2 DNS and UDP

```

PC0
Physical Config Desktop Programming Attributes

Command Prompt
Address: 192.168.1.10
C:\>ping 203.0.113.10

Pinging 203.0.113.10 with 32 bytes of data:

Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127
Reply from 203.0.113.10: bytes=32 time<1ms TTL=127

Ping statistics for 203.0.113.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:

Reply from 192.168.1.101: bytes=32 time=3ms TTL=128
Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Reply from 192.168.1.101: bytes=32 time=4ms TTL=128
Reply from 192.168.1.101: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 2ms

C:\>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:

Reply from 192.168.1.101: bytes=32 time=4ms TTL=128
Reply from 192.168.1.101: bytes=32 time=4ms TTL=128
Reply from 192.168.1.101: bytes=32 time=4ms TTL=128
Reply from 192.168.1.101: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 4ms, Average = 4ms

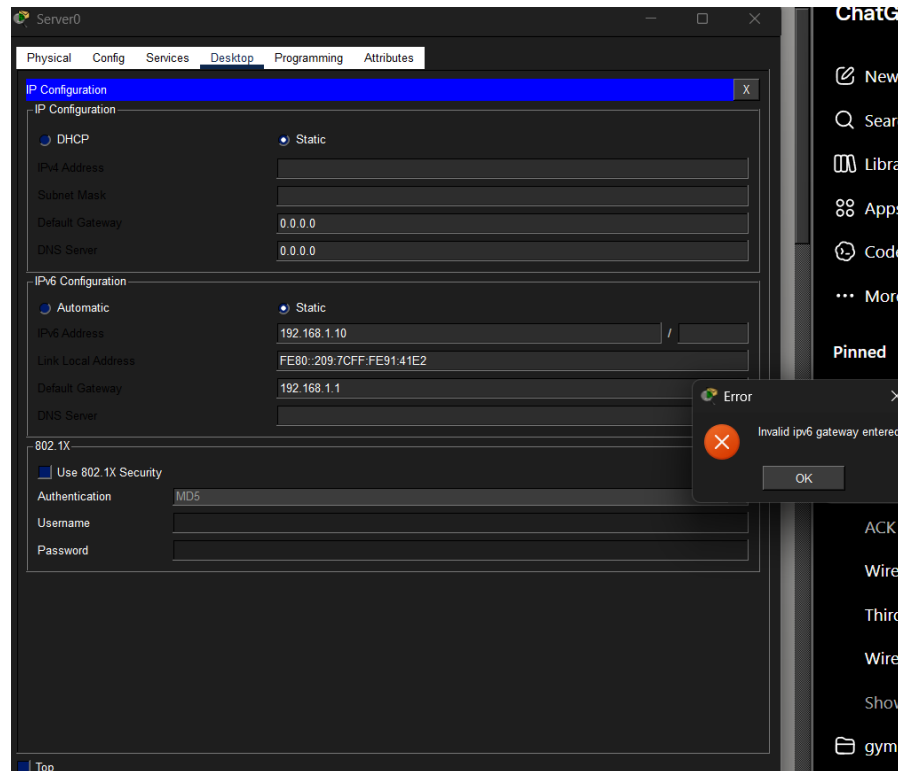
C:\>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:

```

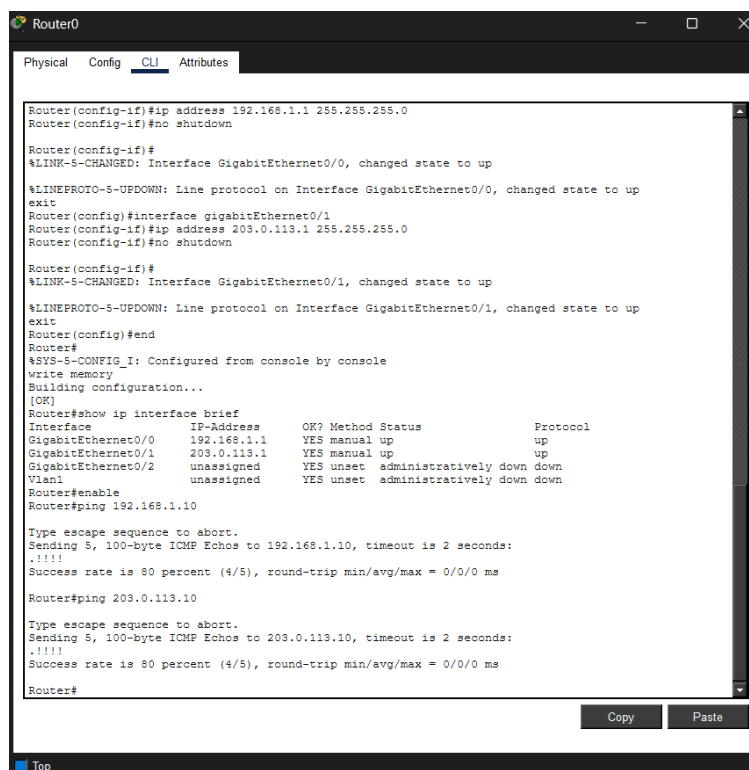
PC0 queried 192.168.1.10 for the A record `www.company.local`. The DNS query used UDP destination port 53, and the server returned 192.168.1.10 from UDP source port 53 to the client ephemeral port. The final `nslookup` returned the expected name and address.

7.3 ARP



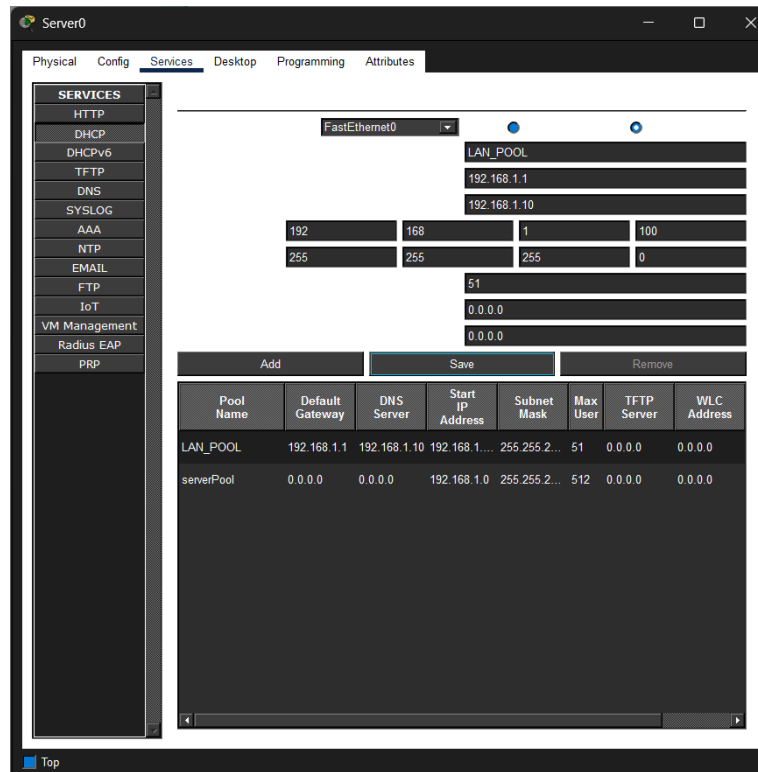
After communication with local devices, `arp -a` on PC0 displayed dynamic mappings for 192.168.1.10 and 192.168.1.101. ARP provided the Layer 2 destination MAC addresses required to transmit Ethernet frames on the LAN.

7.4 ICMP



PC0 successfully pinged the router, internal server, PC1, and external server with 0% packet loss. Simulation Mode showed an ICMP Echo Request from 192.168.1.100 to 192.168.1.101 with message type 8 and an Echo Reply from 192.168.1.101 to 192.168.1.100 with message type 0.

7.5 TCP



Before HTTP data transfer, PC0 created a TCP connection from ephemeral source port 1026 to server destination port 80. The observed sequence represented SYN, SYN-ACK, and ACK, establishing a reliable connection before application data was exchanged.

7.6 HTTP

PDU Information at Device: PC0

OSI Model Inbound PDU Details

Layer 7: HTTP	Layer7
Layer6	Layer6
Layer5	Layer5
Layer 4: TCP Src Port: 80, Dst Port: 1026	Layer4
Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 192.168.1.100	Layer3
Layer 2: Ethernet II Header 0009.7C91.41E2 >> 0005.5ED2.9CD4	Layer2
Layer1	Layer1

1. FastEthernet0 receives the frame.

Challenge Me << Previous Layer Next Layer >>

The browser loaded <http://www.company.local>. The server response used TCP source port 80 and destination port 1026, with source IP 192.168.1.10 and destination IP 192.168.1.100. The custom CSCE 3530 webpage displayed successfully.

7.7 HTTPS and SSL/TLS

PC0

Physical Config Desktop Programming Attributes

Web Browser

< > URL <https://www.company.local> Go Stop

CSCE 3530 Company Network

Welcome to the internal company web server.

DHCP, DNS, HTTP, ICMP, TCP, UDP, and NAT are being demonstrated.

Top

The same page loaded successfully using `https://www.company.local` after HTTPS was enabled on Server0. This demonstrated encrypted web-service access in Packet Tracer.

7.8 NAT Overload (PAT)

After PC0 pinged 203.0.113.10, show ip nat translations displayed ICMP translations from inside local 192.168.1.100 to inside global 203.0.113.1. Multiple ICMP identifiers showed PAT distinguishing individual flows while sharing one public-facing address.

7.9 Routing and Traceroute

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=2ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:

Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Reply from 192.168.1.101: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 203.0.113.10

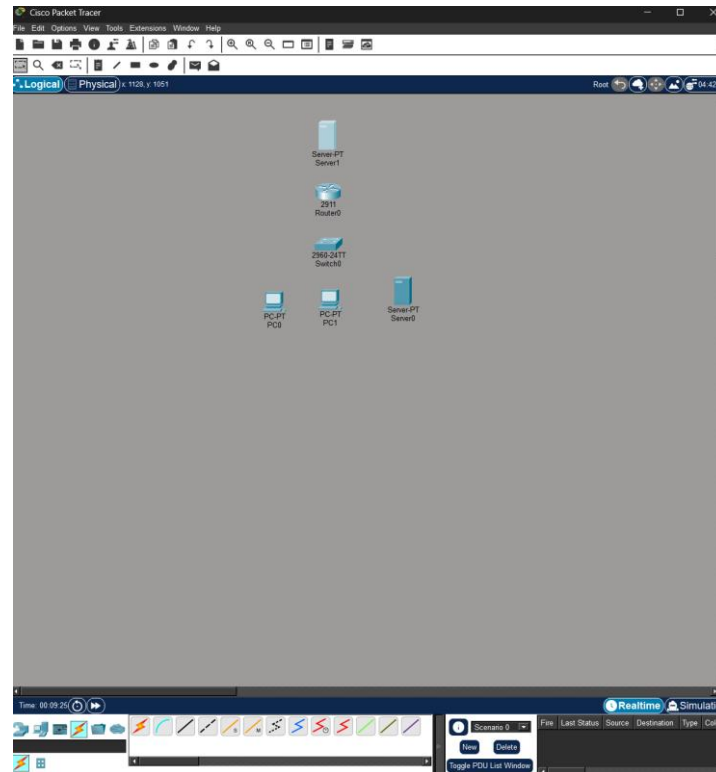
Pinging 203.0.113.10 with 32 bytes of data:
```

tracert 203.0.113.10 completed in two hops: 192.168.1.1 (the internal default gateway) and 203.0.113.10 (the external server). This confirmed correct routing between the LAN and WAN segments.

7.10 Protocol Evidence Summary

Protocol	Observed Source	Observed Destination	Key Evidence
DHCP	0.0.0.0:68	255.255.255.255:67	Discover broadcast; server response 67 -> 68
DNS	192.168.1.100:1033/1034	192.168.1.10:53	A-record query and response
ICMP	192.168.1.100	192.168.1.101	Type 8 request; Type 0 reply
TCP	192.168.1.100:1026	192.168.1.10:80	Three-way handshake
HTTP	192.168.1.10:80	192.168.1.100:1026	Webpage response
NAT/PAT	192.168.1.100	203.0.113.1	Inside local translated to inside global

8. Network Testing Results



Test	Command / Method	Expected Result	Observed Result
Router interfaces	show ip interface brief	Both interfaces up/up	Passed
DHCP PC0	Desktop > IP Configuration > DHCP	192.168.1.100/24	Passed
DHCP PC1	Desktop > IP Configuration > DHCP	192.168.1.101/24	Passed
Gateway connectivity	ping 192.168.1.1	Replies, 0% loss	Passed
Internal server	ping 192.168.1.10	Replies, 0% loss	Passed
PC-to-PC	ping 192.168.1.101	Replies, 0% loss	Passed
External server	ping 203.0.113.10	Replies through router	Passed
DNS	nslookup www.company.local	192.168.1.10	Passed
HTTP	http://www.company.local	Custom page loads	Passed
HTTPS	https://www.company.local	Secure page loads	Passed
ARP	arp -a	Dynamic mappings visible	Passed
Traceroute	tracert 203.0.113.10	Two-hop route	Passed
NAT	show ip nat translations	192.168.1.100 -> 203.0.113.1	Passed

9. Challenges and Solutions

DHCP request initially failed: The DHCP service was initially off. The service was enabled and the client renewed its lease.

Client received 192.168.1.2 with gateway 0.0.0.0: The built-in serverPool had incorrect default values and was selected before the custom pool. The built-in pool was corrected to start at 192.168.1.100 with gateway 192.168.1.1 and DNS 192.168.1.10.

DNS timed out: The DNS service was off and the record name was corrected to `www.company.local`. After enabling DNS, `nslookup` resolved correctly.

Simulation buffer became full: The event list was cleared, Simulation Mode was reset, and protocol filters were limited to only the required protocols. Capture/Forward was used instead of continuous auto-play.

Packet details sometimes showed only Layer 1 and Layer 2: Events at Switch0 were avoided when Layer 3-7 details were needed. PDU windows were opened at PC0, PC1, or Server0.

Traceroute appeared stuck: The test was repeated in Realtime Mode, where it completed successfully.

Packet Tracer rejected `ipconfig /flushdns`: Packet Tracer supports a reduced command set. DNS was re-tested through `nslookup` and a fresh browser request instead.

10. Security and Design Considerations

- NAT hides internal private addresses from the external network, although NAT alone is not a firewall.
- HTTPS was enabled to demonstrate encrypted application-layer communication.
- The internal server combines several services for simplicity. In a production design, DHCP, DNS, and web services may be separated for resilience and security.
- The /24 addressing scheme is simple and supports up to 254 host addresses per segment. A larger organization would use multiple subnets and VLANs.
- The test network does not include access-control lists, switch port security, redundant links, or a dedicated firewall; these are recommended future improvements.

11. Live Demonstration Plan

Time	Demonstration Activity
0:00-1:00	Introduce the project, objectives, and topology.
1:00-2:00	Explain the IP addressing table and router interfaces.
2:00-3:00	Show DHCP settings and both client addresses.
3:00-4:00	Demonstrate ping tests and traceroute.
4:00-5:00	Show DNS record and <code>nslookup</code> .
5:00-6:00	Load the webpage using HTTP and HTTPS.
6:00-7:00	Show ARP table and explain IP-to-MAC mapping.
7:00-8:30	Use Simulation Mode to show ICMP, DNS/UDP, DHCP, and TCP/HTTP events.
8:30-9:30	Show NAT configuration and show <code>ip nat translations</code> .
9:30-10:00	Summarize results and challenges.

12. Conclusion

The project successfully implemented a complete small virtual client-server network in Cisco Packet Tracer. The clients received valid configuration through DHCP, resolved an internal hostname through DNS, communicated with the internal server over HTTP and HTTPS, exchanged ICMP traffic, and reached an external

server through the router. NAT overload translated internal private addresses to the router WAN address, and Simulation Mode provided protocol-level evidence across the OSI model.

The final network met the stated project objectives and demonstrated how addressing, switching, routing, name resolution, transport protocols, web services, and address translation work together in a functional environment. The troubleshooting process also reinforced the importance of service status, correct pool settings, accurate DNS records, protocol filtering, and selecting the correct device when inspecting PDU details.

Appendix A - Complete Router Configuration

```
enable
configure terminal

interface gigabitEthernet0/0
ip address 192.168.1.1 255.255.255.0
ip nat inside
no shutdown
exit

interface gigabitEthernet0/1
ip address 203.0.113.1 255.255.255.0
ip nat outside
no shutdown
exit

access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface gigabitEthernet0/1 overload

end
write memory

show ip interface brief
show ip nat translations
```

Appendix B - Screenshot and Evidence Checklist

- ☐ Final topology with all links green
- ☐ Router show ip interface brief output
- ☐ Internal server static IPv4 configuration
- ☐ DHCP pool settings
- ☐ PC0 DHCP address 192.168.1.100
- ☐ PC1 DHCP address 192.168.1.101
- ☐ Successful ping results
- ☐ DNS A record and nslookup result
- ☐ HTTP webpage loaded by hostname
- ☐ HTTPS webpage loaded by hostname

- ☐ NAT translation table
- ☐ ICMP Echo Request and Echo Reply PDU details
- ☐ DNS query/response over UDP PDU details
- ☐ TCP connection initiation, server response, and acknowledgment
- ☐ HTTP response PDU details
- ☐ DHCP Discover PDU details
- ☐ DHCP server response to PC1
- ☐ ARP table output
- ☐ Traceroute to 203.0.113.10

End of Report